

A Situational Speech Synthesizer for Yorùbá:

System Design, Phonological Rule Architecture, and Orthographic Extensions for Contour Tones

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ABSTRACT

We present TTSYoruba, a rule-based concatenative diphone speech synthesizer for Yorùbá, deployed at online as part of the YorubaName.com open dictionary of Yorùbá personal names. The system takes tone-marked Yorùbá text as input and produces audio output by applying a hand-crafted phonological rule system to a recorded inventory of 651 diphone units spanning five tonal variants of every consonant-vowel combination in the language. We describe the phonological architecture of the system in detail, including our complete tonal file-selection logic, our treatment of the three-way nasal disambiguation problem (oral /n/, nasalized vowel, and syllabic nasal), and the derivation of contextual rising and falling tones from level-tone input. We also present, as an orthographic contribution, the adoption of the caron (ˇ) and circumflex (^), which are symbols with prior standing in Yorùbá phonological transcription, as standard single-vowel contour tone markers, integrated into the TTS normalization pipeline and the WriteYoruba keyboard input tool. The system's performance was evaluated through a listener study (N=50), with detailed results on Mean Opinion Scores (MOS) presented in Section 6.

Keywords: *Yorùbá, text-to-speech, low-resource languages, diphone synthesis, contour tones, African language NLP, rule-based synthesis*

1. INTRODUCTION

Yorùbá is one of the few African languages with a standardized writing system that compulsorily encodes lexical tone directly in its orthography. High tone uses the acute accent [ó], low

tone uses the grave accent [ò], and mid tone is usually left unmarked [o]. This feature makes Yorùbá unusually tractable for rule-based text-to-speech (TTS) synthesis: the graphemic input already carries the phonological information that most TTS systems must infer from probabilistic models.

In 2015, the Yorùbá Names Project was established to build a free, open database of Yorùbá personal names with morpheme breakdowns, tone markings, meanings, and pronunciation guides (Túbòsún, 2020). A core engineering goal was automated pronunciation. Rather than manually recording each name's audio, which is an approach that does not scale as the database grows through community contribution, the project required a synthesis engine capable of generating correct pronunciation for any correctly tone-marked input name.

This paper describes the resulting system, TTSYoruba, and makes the following contributions:

- A 651-unit diphone corpus covering all phonologically predicted syllable-tone combinations in Yorùbá, recorded under consistent acoustic conditions.
- A complete tonal file-selection rule system specifying, for every grapheme-tone combination, which diphone file is selected and under what tonal environment conditions, published here in full for the first time.
- A three-way nasal disambiguation system distinguishing oral /n/, nasalized vowels, and syllabic nasals, with the decision logic made explicit.
- A new diacritical convention, Unicode caron (ˇ) for rising contour and circumflex (^) for falling contour on single vowels, to resolve a class of previously unsynthesizable names, with integration into both the TTS pipeline and the WriteYoruba keyboard tool.

The system is deployed at TTSYoruba.com and integrated into the YorubaName.com dictionary, where it has generated audio for almost 10,500 name entries since launch.

While components of this system have been in iterative development and use since 2017, this paper serves as the first formal, comprehensive documentation of the complete tonal file-selection

architecture and the nasal disambiguation logic, intended to establish these specifications as a public record and a standard for future Yorùbá speech technology development.

2. BACKGROUND AND RELATED WORK

2.1 Phonological properties of Yorùbá relevant to TTS

Yorùbá has three contrastive level tones: high (H), mid (M), and low (L), with the syllable being the primary tone-bearing unit. Tones are marked on vowels and, where applicable, on syllabic nasals (e.g. *ń*, *ń*). The language has seven oral vowels (a, e, ẹ, i, o, ọ, u) and eighteen consonants (b, d, f, g, gb, h, j, k, l, m, n, p, r, s, ʃ, t, w, y). Syllable structure is predominantly CV, with additional types comprising standalone vowels (V) and syllabic nasals (N). (*see Awobuluyi, 1994, for a history of orthographic development*).

The three tones do not behave symmetrically under phonological pressure. Pulleyblank (2004) demonstrates that High tones are the most stable, most resistant to deletion and most likely to be retained when tones compete, while Mid tones are the least stable, most readily lost, and incapable of participating in lexical contour formation. Low tones are intermediate. This asymmetry is directly encoded in the file-selection logic of the TTSYoruba system through a five-file inventory (*_l*, *_m*, *_h*, *_f*, *_r*) that includes rising and falling contour variants for High and Low tones only. No *_mf* or *_mr* file exists, and no Mid-tone syllable triggers contextual rising or falling assignment. That design reflects the phonological facts: Mid tones do not form lexical contours in Yorùbá.

Beyond the three level tones, phonological processes yield two contour tone types: a rising tone (L+H), realized when a high-tone syllable directly follows a low-tone syllable in a tonal sandhi environment; and a falling tone (H+L), realized when a low-tone syllable directly follows a high-tone syllable. These contour realizations are predictable from the surrounding tonal context and are encoded in the diphone inventory as separate units (files suffixed *_r* and *_f* respectively). (See also Ọlátúbòsún 2012, pp. 13–15, for experimental data on the perceptual salience of rising and falling tones for non-native speakers.)

The use of the caron and circumflex as contour tone symbols has precedent in Yorùbá phonological scholarship: Olmsted (1951) employs them in the first major international phonemic analysis of the language, and Oyètádé (1988) uses them in an autosegmental treatment of Yorùbá tonal processes. Section 5 describes how this notation, previously confined to phonological transcription, has been extended to orthographic practice and integrated into the TTSYoruba synthesis pipeline.

2.2 Prior work on Yorùbá TTS

Ọdẹjọbí et al (2009) proposes a syllable-based duration model for Yorùbá TTS prosody. Ọrẹ (2014) develops a grapheme-to-phoneme conversion system. Iyanda et al. (2014) present a phonemic description framework for Yorùbá speech. None of these systems has been deployed as a free and publicly accessible tool. TTSYoruba extends this prior work by providing a complete, deployed system whose rule architecture is documented in sufficient detail to support replication.

2.3 The case for rule-based synthesis in a low-resource setting

Neural TTS systems have achieved high naturalness on well-resourced languages but require hundreds of hours of annotated speech data that do not yet exist for Yorùbá. Rule-based concatenative synthesis requires only a small diphone corpus, produces interpretable and auditable behavior, and offers consistent output for the target domain of personal names, which are typically short, clearly articulable, and lack the prosodic complexity of connected speech. These properties make rule-based synthesis a principled choice for the current application.

3. DIPHONE CORPUS AND TONAL FILE ARCHITECTURE

3.1 Corpus construction

All diphones were recorded by the first author in a home studio environment using an Audio-Technica AT2020+ microphone and Audacity 2.1.1 at 44.1 kHz stereo. Individual diphone segments were cut and normalized to consistent amplitude levels. The inventory is organized as audio files named by a phoneme-tone key convention:

File suffix	Tone	Environment	Example file	Example word
_l.wav	Low	Default; not after H or R tone	ba_l.wav	<u>Bà</u> bá (bà-)
_m.wav	Mid	Anywhere	ba_m.wav	<u>Ba</u> lógun (ba-)
_h.wav	High	Default; not after L or F tone	de_h.wav	A <u>dé</u> (-dé)
_f.wav	Falling (H→L)	After H or R tone; or circumflex mark (â)	boo_f.wav	Adé <u>bò</u> wálé (-bò-)
_r.wav	Rising (L→H)	After L or F tone; or caron mark (ǎ)	wa_r.wav	Adébò <u>wá</u> lé (-wá-)

Table 1. Tonal file-suffix conventions and their environments. The five-way tonal distinction (_l, _m, _h, _f, _r) applies to every CV diphone in the inventory.

Open vowels ɛ and ɔ are represented by doubling the vowel letter in the file key: “bẹ” maps to *bee_h.wav*, “bò” maps to *boo_l.wav*, and so on. The grapheme ‘ş’ and the digraph ‘sh’ share the same audio files (sha_, she_, etc.), as they represent the same phoneme /ʃ/ in Yorùbá orthography.

3.2 Inventory size

Category	Base units	Tone variants	Subtotal
CV diphones (18 consonants × 7 vowels)	126	5 per pair	630
Word-initial standalone vowels	7	3 (L, M, H)	21

Category	Base units	Tone variants	Subtotal
Total recorded units	—	—	651

Table 2. Summary of the TTSYoruba diphone inventory.

The five tonal variants per CV pair arise from the three level tones plus the two contextually conditioned contour realizations. Standalone vowels take three variants because they occur word-initially and do not participate in the same post-consonant tonal sandhi environments as CV syllables.

4. PHONOLOGICAL RULE SYSTEM

The rule system converts a tone-marked input string into a sequence of diphone file keys, which are looked up and concatenated to produce the audio output. The pipeline applies four modules in sequence: (1) Unicode normalization, (2) syllabification and nasal disambiguation, (3) tonal file selection, and (4) diphone concatenation. We describe each in turn.

4.1 Unicode normalization

Input text is normalized to Unicode NFC (Canonical Decomposition, followed by Canonical Composition) before processing, ensuring that diacritic-bearing characters are represented as precomposed forms regardless of the input encoding path. This is essential because Yorùbá text may arrive from different keyboards and applications with different Unicode normalization states.

4.2 Tonal file selection

The core rule for file selection is straightforward: the file suffix is determined by the combination of the graphemic tone marking on the current syllable and the tone of the immediately preceding syllable. The full rule set is captured in Table 1 above. Two additional cases require elaboration:

Contextual rising tone: A grapheme marked high tone (acute accent) is assigned the rising (*_r*) diphone file when it immediately follows a syllable assigned a low or falling tone. Example: in Adébòwálé, the syllable *-wá-* follows the falling-tone *-bò-*; the file selected is *wa_r.wav*, not *wa_h.wav*. This captures the phonological process by which a high tone is phonetically realized as a rising contour in this environment.¹

Contextual falling tone: A grapheme marked low tone (grave accent) is assigned the falling (*_f*) diphone file when it immediately follows a syllable assigned a high or rising tone. Example: in Adébòwálé, the syllable *-bò-* follows the high-tone *-dé-*; the file selected is *boo_f.wav*.

Table 3 traces the complete file-selection sequence for the name Adébòwálé:

Input syllable	Preceding tone	File selected	Tone assignment
a-	Word-initial	a_m.wav	Mid (default initial)
-dé	After mid	de_h.wav	High (diacritic)
-bò-	After high	boo_f.wav	Falling (low after high → <i>_f</i>)
-wá-	After falling	wa_r.wav	Rising (high after falling → <i>_r</i>)
-lé	After rising	le_h.wav	High (diacritic, confirmed by environment)

Table 3. File-selection trace for Adébòwálé, illustrating contextual rising and falling tone assignment.

4.3 Nasal disambiguation

¹ This rule implements, computationally, the output of the MAX[H] and MAX[L] faithfulness rankings that Pulleyblank (2004) argues govern Yorùbá tonal stability: a High tone is preserved as rising rather than lost, because losing it entirely would incur a MAX[H] violation more costly than the contour realization.

The nasal consonant /n/ (and by extension /m/) is the most structurally ambiguous element in Yorùbá orthography for TTS purposes. It may function in three distinct phonological roles, each mapping to a different diphone file:

1. Oral onset: /n/ begins a CV syllable. Example: Adénólá → a-dé-nó-lá. File: *n_o_o_h.wav* for “nó”.
2. Nasalization marker: /n/ following a vowel nasalizes that vowel and is not itself a separate syllable. Example: Sànyà → /sǎ/-/ja/. File: the preceding vowel-consonant pair is replaced by the corresponding nasal diphone (e.g. *san_l.wav*).
3. Syllabic nasal: /n/ or /m/ constitutes a complete syllable bearing its own tone. Example: Àńjólá → à-ń-jò-lá. File: *n_r.wav* (rising, because it follows a low-tone syllable).

The disambiguation rules, derived from systematic analysis of the Yorùbá name corpus, are as follows:

Rule N1 (syllabic, explicit). If /n/ or /m/ carries a tone diacritic (e.g. ń, ò, ń, ò), treat as syllabic. File: *n_h.wav*, *n_l.wav*, *m_h.wav*, etc.

Rule N2 (nasal vowel, vowel class). If /n/ is preceded by the vowels /i/ or /u/ in mid-word position, and followed by a consonant, treat /n/ as a nasalization marker and assign the nasal diphone to the preceding vowel-onset pair. Example: Àdùntólá → à-dùn-tò-lá, because /u/ precedes /n/ which is followed by /t/. The morphological structure places the boundary after dùn.

Rule N3 (oral onset, default). If /n/ is preceded by /o/, /e/, /o/, or /e/ in mid-word position and followed by a vowel, treat /n/ as an oral onset consonant for the following syllable. Example: Adénólá → a-dé-nó-lá.

Rule N4 (word-final nasalization). Any word-final /n/ nasalizes the preceding vowel.

The three-way classification underlying these rules (i.e: *n* as consonant onset, *n* as syllabic nasal, and *n* as nasality marker on a preceding vowel) is given a formal linguistic treatment in Ajiboye (2020), who

notes that the convention of using *n* to mark nasal vowels rather than the diacritic [̃] was a deliberate decision of the 1974 Orthography Committee made on practical typing grounds. The computational consequence of that decision, that is, the ambiguity that rule-based TTS must resolve, is what Rules N1–N4 address.

Table 4 illustrates these rules across representative names:

Name	Syllabification	Rule class applied	Notes
Sànyà	sā – ja	Nasal vowel	/n/ after /a/ before consonant → nasalizes vowel
Sám̀bà	sá – m̃ – bà	Syllabic nasal	/m/ before bilabial → syllabic
Sángo ²	sā – ŋ – go	Mixed	Nasal vowel + syllabic /ŋ/
Àńjọlá	à – ṇ – jọ – lá	Syllabic nasal	Tone mark on /n/ → <i>n_r.wav</i>
Adénọ́lá	a – dé – nọ – lá	Oral /n/	/n/ between non-high vowels → oral
Ìtànìfẹ	ì – tã – ì – fẹ	Nasal vowel	/an/ at morpheme boundary: itàn + ifẹ
Àdùnọ́lá	à – dùn – ọ – lá	Nasal diphone	/u/ precedes /n/ → <i>dun_f.wav</i> group

² The place name, which is different from the name of the deity, should ordinarily be written as *Sánńgo*, if accounting for all the variations of *n* in the word were the object of the spelling, but this is rarely the case, due to these occasional arbitrariness in Yorùbá spelling. It's the same with the name “Ògúndé” (which is pronounced Ògúnndé).

Table 4. Nasal disambiguation examples showing the three rule classes and their outputs.

An acknowledged limitation is the morpheme-boundary case, like in *Ìtànìfẹ́* (morphological decomposition: *ìtàn* + *ìfẹ́*). Rule N3 as stated incorrectly parses the second syllable as /ta/ + /ni/ rather than /tǎ/. Even a name like *Adùnlá* (which admits a rare alternative pronunciation *A-dù-nó-lá* in some peculiar instances) presents an apparent exception. However, the most widely attested form is the nasal variant, and Rule N3 correctly handles it. Resolving the likely many more exceptions in this class requires either a morphological parser or a manually curated exception lexicon. The current system handles such names by lexical override when they are identified as problematic.

5. ORTHOGRAPHIC EXTENSION: ENCODING CONTOUR TONES ON SINGLE VOWELS

5.1 The problem

The contextual rising and falling tone rules in Section 4.2 handle contour tones that arise from the tonal environment of a syllable within a correctly tone-marked word. A separate and more difficult class of cases arises for personal names whose conventional spellings omit the vowel gemination that standard Yorùbá orthography would use to represent a contour tone within a single syllable.

The standard solution is geminated vowel (e.g. vowel doubling), where a name like “Nike” is written with a doubled vowel e.g. “Níkèè” or “Nîkè”, each instance carrying one component tone of the rising contour. Native speakers do not typically use this spelling because established names circulate in forms that omit this doubling, either because they entered print before systematic tone-marking conventions were established, or because their bearers prefer the inherited spelling. A TTS system that processes names as written cannot synthesize contour tones for this class without either distorting the spelling or leaving the tone unrepresented. A fuller linguistic and orthographic treatment of this problem, including its historical context and the formal equivalences between contour-marked and geminated forms, is presented in Túbòsún (forthcoming).

Native speakers who encounter geminated contour orthography outside of linguistic or pedagogical contexts often perceive it as erroneous or unusual: in the MOS listening study reported in Section 6, one participant spontaneously noted that the spelling *Akẹ̀kẹ̀ọ̀*, the standard geminated form of the word for “student”, was “off,” while the caron equivalent *Akẹ́kẹ́* elicited no such comment. This asymmetry illustrates the practical cost of geminated notation: even when phonologically accurate, it may trigger unfamiliarity responses that interfere with the perception of correctness.

This problem was identified but unresolved in the first version of this system. We now present a solution.

5.2 Proposed convention

We propose encoding contour tones on single vowels using two Unicode diacritics:

- Caron (ˇ), yielding e.g. *ǎ, ě, ě, ĭ, ǫ, ǫ, ů*: to encode a rising (L+H) contour on a single vowel.
- Circumflex (^), yielding e.g. *â, ê, ê, î, ô, ô, û*: to encode a falling (H+L) contour on a single vowel.

These marks are already defined in the Unicode Latin Extended range, are supported in common fonts, and can be combined with the subscript dot used for the open vowels *ẹ* and *ọ*. They map to the existing *_r* and *_f* diphone files respectively, so no new audio recordings are required. The formal equivalences are:

ǎ ≡ *àá* (rising contour on /a/)

â ≡ *àà* (falling contour on /a/)

ě ≡ *èé* (rising contour on /e/)

ê ≡ *èè* (falling contour on /e/)

ě ≡ *ẹẹ* (rising contour on /ẹ/)

ê ≡ *ẹẹ* (falling contour on /ẹ/)

ĩ ≡ *ìí* (rising contour on /i/)

î ≡ ì (falling contour on /i/)

ố ≡ òó (rising contour on /o/)

ô ≡ òò (falling contour on /o/)

ỗ ≡ ồỗ (rising contour on /o/)

ộ ≡ ồò (falling contour on /o/)

ũ ≡ ùú (rising contour on /u/)

û ≡ ùù (falling contour on /u/)

These equivalences are already reflected in the rulebook that governs the TTS engine. Entry 1136b, for instance, specifies that the caron form “răn” (as in Arómọlárăn) maps to *roon_r.wav*, exactly as the geminated form *ràán* does. Entry 260b specifies that “kě” (as in Adénîkẹ) maps to *kee_r.wav*. The convention therefore requires no changes to the diphone inventory; only a normalization step in the preprocessing pipeline.

5.3 Examples

Common spelling	Full tonal form	Contour-marked form	TTS file key (contour syllables)
Aromọlaran	Arómọláraán	Arómọlárăn	<i>roon_r.wav</i> (răn)
Ijerisi	Ìjẹ̀rìsì	Ìjẹ̀rĩsì	<i>ri_r.wav</i> (rĩ)
Adenikẹ	Adénîkẹ	Adénîkẹ	<i>kee_r.wav</i> (kẹ)
Ajafẹ̀tọ	Ajàfẹ̀tò	Ajàfẹ̀tọ	<i>too_r.wav</i> (tọ)
Adebiyi	Adébíyí	Adébíyĩ	<i>yi_r.wav</i> (yĩ)

Common spelling	Full tonal form	Contour-marked form	TTS file key (contour syllables)
Alawiye	Aláwíyé	Alawǎyé	<i>wi_r.wav</i> (wǐ)
Ayotomi	Ayòótómi	Ayǒtómi	<i>yoo_r.wav</i> (yǒ)
Oyedeji	Oyèédèjì	Oyědèjì	<i>ye_r.wav</i> (yě)

Table 5. Eight Yorùbá personal names showing the common spelling, full geminated tonal form, proposed contour-marked form, and the TTS diphone file key for the contour-bearing syllable.

5.4 Implementation

Keyboard input. The WriteYoruba keyboard software (macOS and Windows, WriteYoruba.com, first released 2016) was extended with Option-key and Alt-key combinations for the caron and circumflex on all Yorùbá vowels, including the open variants ẹ and ọ.

TTS normalization. A preprocessing normalization step was added to the TTSYoruba pipeline: before grapheme-to-phoneme conversion, caron-marked and circumflex-marked vowels are expanded to their geminated equivalents, after which processing continues normally. For example:

Arómọlárǎn → normalize → Arómọláràán → G2P → [*roon_r.wav* ...]

Caron normalization is currently active for all vowels. Circumflex support for the full open-vowel set is in progress. The practical result is input-form equivalence: whether a user enters Níkẹ́ẹ́, Níkẹ́, or Níkẹ́, the synthesizer produces the same rising-contour audio output.

5.5 Historical note

The use of caron and circumflex in informal Yorùbá handwriting is not a modern invention. It is an informal adaptation of a notation with formal phonological standing from at least 1951. Olmsted (1951) used the caron and circumflex as tone contour symbols in the first major phonemic study of Yorùbá published in an international linguistics journal; Oyètádé (1988) employed the same notation

in a formal autosegmental analysis. What was missing was a path from phonological transcription into standard orthography and computer input, which is what this paper addresses.

The 1974 Yorùbá Orthography Committee (Joint Consultative Committee on Education, 1974) acknowledged the problem indirectly: Item 11 of the revised orthography discontinued the tilde (~), which had served as a single-symbol contour notation, replacing it with vowel gemination. The tilde's discontinuation removed the only contour marker from standard orthography and left gemination as the sole available representation. The caron and circumflex proposal reinstates single-symbol contour notation using Unicode-compatible characters that are unambiguous, typeable, and combinable with the existing diacritic system.

The structural depth of this gap extends beyond the Roman orthographic tradition, as Ogunbiyi (2003) documents for Yorùbá *Ajamī* writers (primarily in Muslim communities from the pre-colonial period onward) who had previously used the Arabic script to write Yorùbá. They faced the same unresolved problem with tones, contour tones especially could not be adequately represented using the available diacritical resources of either the Arabic or Roman writing systems (pp. 92–93, 97). The caron and circumflex convention described in this section thus addresses a problem that has resisted solution across two orthographic traditions over nearly two centuries.

6. EVALUATION

We conducted a Mean Opinion Score (MOS) listening study to assess the naturalness and intelligibility of TTSYoruba output across four structurally distinct name categories. The study also addresses a question raised by Fajobi and Akomolafe (2019), whose investigation of anglicised Yorùbá personal names found that bearers systematically apply Yorùbá tonal patterns even to structurally anglicised forms, suggesting that tonal accuracy, not merely segmental intelligibility, is the relevant perceptual criterion for a Yorùbá name synthesiser. This motivated our inclusion of two separate rating dimensions.

6.1 Study design and participants

The study was administered as a browser-based listening instrument hosted at <https://www.writeyoruba.com/ttstest>. Fifty participants were recruited from the YorubaName.com contributor community and through the project's social media channels; all were at least 18 years of age. Data collection ran over a six-day period (26 June – 2 July 2026). Demographic information collected at the outset included age range, regional Yorùbá variety, self-rated Yorùbá proficiency on a five-point scale, and listening-device type. Responses were anonymous; participants could optionally provide an email address solely to prevent duplicate submissions. Each participant completed a short unscored practice trial before the study proper.

The sample was strongly weighted toward native and near-native speakers (43 of 50, 86%), with the remainder rating themselves as advanced (3), intermediate (3), or beginner (1). Participants spanned the 18–24 (n=4), 25–34 (n=24), 35–44 (n=14), 45–54 (n=5), and 55–64 (n=3) age brackets, and reported a broad range of regional varieties including Standard Yorùbá, Ọ̀yọ́, Èkìtì, Ìjẹ̀bú, Ìbòlò, Ìgbómìnà, Òndó, and Èkó. The median completion time was 4.8 minutes.

A majority of participants listened via phone speaker (30, 60%), with the remainder using earphones (12, 24%), external speakers (3, 6%), laptop speakers (4, 8%), or over-ear headphones (1, 2%). As phone speakers may attenuate acoustic cues relevant to tonal perception, this distribution likely places a conservative floor on the naturalness scores relative to a controlled headphone-listening condition.

6.2 Stimuli

The stimulus pool comprised synthesised audio generated by TTSYoruba for personal names drawn from four categories corresponding to the system's principal phonological domains:

- **Category A (Standard tonal):** names requiring only the standard three-level tone file-selection rules and no contextual contour assignment (e.g. Kòlá, Yẹ́mísí, Títílọ́lá, Bájomọ etc).
- **Category B (Syllabic nasal):** names containing syllabic nasals bearing their own tone, engaging the nasal disambiguation rules of Section 4.3 (e.g. Adébanjọ, Ifátúnmbí, Gbénga, Pónlé, etc).
- **Category C (Geminated contour):** names with rising or falling contour tones represented through vowel gemination in standard orthography (e.g. Arómọláraán, Adéníkẹẹ, Aláwíiyé, Oníkàán).
- **Category D (Caron/circumflex contour):** the same underlying names as Category C, written using the single-vowel contour notation proposed in Section 5 (e.g. Arómọlárăn, Adéníkẹ, Aláwíiyé, Oníkà).

Adéníkẹ, Aláwíyẹ, Oníkàn, etc). This category directly tests the equivalence claim of Section 5.2.

Each participant heard ten stimuli: two random names from Category A, two from Category B, two matched C/D pairs (four items, ensuring each session contained both the geminated and caron forms of the same underlying name), and two additional items from any category. Presentation order was randomised per participant, and paired C/D items were not presented adjacently. An attention-check item (a synthesis of Kólá) was embedded at a random non-initial, non-final position in each session; sessions rating it 1 or 2 on both dimensions simultaneously would have been excluded.

6.3 Rating instrument

For each stimulus, participants rated two dimensions on a five-point scale following the ITU-T P.800 convention: naturalness (1 = completely unnatural; 5 = completely natural) and intelligibility (1 = incomprehensible; 5 = fully intelligible). Rating controls were disabled until the participant had played the audio, ensuring no item was rated unheard; participants could replay each stimulus freely, and the replay count was logged. An optional free-text comment field was provided per stimulus. Participants were encouraged but not required to use headphones or earphones.

6.4 Results

Fifty participants provided 500 scored ratings (excluding attention-check items). All 50 sessions passed the attention check (100% pass rate), and mean replay count was low and uniform across categories (1.7–2.0), indicating that stimuli were generally rated on a single or second listen. Table 6 reports mean naturalness and intelligibility by category.

Name category	Naturalness	Intelligibility	Example stimuli
A: Standard tonal	4.15 (0.92)	4.60 (0.68)	Kólá, Títílọlá, Ọpẹyẹmí...
B: Syllabic nasal	3.90 (0.97)	4.48 (0.72)	Bánkẹ, Adébánjọ, Ifátúnmbí...
C: Geminated contour	4.10 (1.01)	4.52 (0.80)	Arómọláraán, Adéníkẹ...
D: Caron/circumflex	4.14 (1.02)	4.60 (0.69)	Arómọlárăn, Adéníkẹ...
Overall	4.08 (0.98)	4.55 (0.72)	all categories

Table 6. Mean Opinion Scores (1–5) by name category; standard deviations in parentheses. Overall N = 500 scored ratings from 50 participants.

Intelligibility was uniformly high across all four categories (4.48–4.60), indicating that TTSYoruba reliably produces understandable output regardless of phonological structure. Naturalness scores were more moderate (3.90–4.15), consistent with the known characteristics of concatenative diphone synthesis. A one-way ANOVA found no significant differences among the four categories on either dimension (naturalness: $F(3,496)=1.79$, $p=.148$; intelligibility: $F(3,496)=0.84$, $p=.472$). Syllabic-nasal names (Category B) received the lowest scores on both dimensions, a pattern that was consistent across the collection period and is discussed below.

The central comparison of this study concerns Categories C and D: the geminated and caron/circumflex notations for the same underlying contour-toned names. Because each session presented both members of two matched pairs, we compared them using paired-samples t-tests over 100 matched observations. Neither dimension showed a significant difference: naturalness $t(99)=-1.03$, $p=.307$, $d=0.10$; intelligibility $t(99)=-1.65$, $p=.101$, $d=0.17$. The two notations are therefore perceptually equivalent in synthesis, supporting the claim of Section 5.2. We note that the small numerical differences that did obtain favoured the caron form on both dimensions, but neither approached significance in the full sample.

Qualitative comments reinforce and extend these findings in three respects. First, the dominant theme across all categories, which appeared in a substantial share of free-text responses, was inter-syllable pause duration: participants repeatedly described the synthesis as “too slow,” as pronounced “at syllabic level and not natural flow,” and as resembling “teaching or textbook” speech directed at non-native learners. This reflects a known limitation of concatenative diphone synthesis and is consistent with the moderate naturalness scores observed throughout. Second, multiple participants independently observed for different name pairs that the geminated and caron forms sounded “exactly” alike; one wrote of the caron form *Lófě* that it “sounds exactly like the previous one with the double ee,” providing qualitative corroboration of the statistical equivalence result. Third, and of direct relevance to Section 5.1, one participant described the geminated spelling *Akẹ̀kẹ̀* as “off” while rating its caron equivalent *Akẹ́kẹ́* as “this is it — good,” a spontaneous native-speaker reaction illustrating the perceptual cost of geminated notation in everyday contexts.

Because a subset of participants attended to text–audio correspondence (including orthographic form) in addition to audio quality, we conducted a sensitivity analysis. Ten sessions were identified from their comments as attending to spelling or tonal-marking correspondence. Excluding these sessions shifted every category mean by at most 0.10 points and did not alter the outcome of the C/D equivalence tests

(naturalness $p=.783$; intelligibility $p=.211$, $n=80$ pairs). These sessions are therefore retained in the reported analysis.

Taken together, the evaluation supports three conclusions. TTSYoruba produces highly intelligible Yorùbá across all four name categories; its naturalness is moderate and limited primarily by the inter-syllable discontinuities characteristic of concatenative synthesis; and the caron/circumflex notation proposed in Section 5 is perceptually indistinguishable from the established geminated notation, incurring no cost in either naturalness or intelligibility. To our knowledge this is the first published listener evaluation of a Yorùbá text-to-speech system, and the scores reported here establish a baseline against which future neural or hybrid Yorùbá synthesis systems may be compared.

7. KNOWN LIMITATIONS AND FUTURE WORK

7.1 Morpheme-boundary nasal errors

The nasal disambiguation rules fail for a subset of names at morpheme boundaries where a nasalized vowel is immediately followed by a vowel-initial morpheme. “Ìtànìfẹ́” (ìtàn + ìfẹ́) is the prototypical case. Resolution requires morphological segmentation or an expanded lexical exception list.

7.2 Prosodic modeling for multi-word utterances

The system was designed for single-name synthesis and avoids the prosodic challenges of connected speech (downstep, downdrift, high-tone raising; cf. Laniran and Clements, 2003). Extending the system to sentences requires a separate prosodic model and pitch modification at synthesis time, beyond the scope of concatenative diphone synthesis. The rule, in principle, could be extended to sentence-level input, but the constraint there is the nature of pre-recorded diphone concatenation itself. Without on-the-fly pitch contour modification, the system cannot model the gradual pitch declination (downdrift), register-resetting (downstep), or boundary tone effects that characterize natural connected Yorùbá speech.

Beyond the current architecture, the phonological rule system documented in this paper is designed to function as a language-independent preprocessing layer, creating a direct path toward a

hybrid architecture in which the rule system serves as a text normalization front-end for a neural TTS model. The caron and circumflex forms are expanded to their geminated equivalents, nasal disambiguation is resolved, and contextual contour assignment is applied before the normalized input is passed to a neural acoustic model. Such an architecture would preserve the linguistic precision of the rule-based layer while delegating acoustic modeling, including pitch contour, coarticulation, and prosodic phrasing, to a learned model trained on a naturalistic Yorùbá speech corpus such as ÌròyìnSpeech (Ògúnṛẹ́mí et al., 2024). We regard this hybrid approach as the most promising near-term direction for improving naturalness in Yorùbá TTS, and the rule documentation provided here is intended partly to support that future work.

7.3 Input literacy dependency

The system requires correctly tone-marked input. Users who cannot produce tone marks cannot use the system without assistance. These are a significant population, given limited formal Yorùbá literacy instruction and adoption. The WriteYoruba keyboard tool partially addresses this barrier. Integration with an upstream Automatic Diacritic Restoration (ADR) system (cf. Ọ̀rìfẹ́ et al., 2020) would address it more fully.

The architecture described here is specific to languages in which lexical tone is marked obligatorily in the standard orthography. Igbo and several other Nigerian tonal languages do not mark tone in their conventional written forms, meaning that a rule-based concatenative system of this type cannot be directly applied without first solving the upstream problem of automatic tone restoration from unmarked text. For such languages, a statistical or neural approach, along with the construction of a tone-marked Igbo corpus, would be a necessary prerequisite, rather than an extension of the diphone rule system described here.

7.4 Circumflex completion

Circumflex support for falling contour tones on open vowels (ê, ô) is partially implemented. Full support across all vowel types is planned for the next release.

7.5 Upstep

A tonal phenomenon termed "upstep", in which a high tone is realized at a significantly elevated pitch with a raising effect on contiguous tones, as in *ìsekúse*, is documented in Ọlátúbòsún (2012, pp. 14–15) but is not currently modeled in the TTSYoruba system. Incorporating upstep into the rule architecture is a direction for future work.

8. CONCLUSION

We have described TTSYoruba, the first publicly deployed speech synthesizer for Yorùbá, and documented its phonological rule architecture in sufficient detail to support evaluation and replication. The system's diphone file-selection logic, nasal disambiguation rules, and contour tone handling are all made explicit here for the first time. The new diacritical convention for contour tones, which extends the system's coverage to a class of names that could not previously be correctly synthesized, is a contribution that goes beyond this system: it offers any orthographic, computational, or pedagogical tool working with Yorùbá text a Unicode-compatible, keyboard-accessible way to represent rising and falling contour tones on single vowels without altering established spellings.

The system demonstrates that low-resource language TTS development and linguistic description are mutually reinforcing: building the synthesizer required formalizing phonological rules that had not previously been computationally specified, and the synthesis failures that emerged from deployment drove the orthographic innovation described in Section 5. We offer this record of iterative development as a practical model for researchers working on speech technology for other low-resource tonal languages.

A planned subsequent version will replace the current single-speaker (male) diphone inventory with a recording by a female speaker, using an identical file-naming convention and rule architecture.

This substitution would require no changes to the phonological rule system, demonstrating the modularity of the acoustic and linguistic layers of the design.

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